

qPCR readout

Introduction

Based on in house protocol

Reverse Transcription based on [High Capacity cDNA Reverse Transcription Kit](#)

Materials

- › PBS
- › Dry ice
- › RNeasy Kit
- › DPEC water
- › Qubit reagents
- › 10X RT Buffer
 - › RT box -20C
- › dNTP Mix (100 mM)
 - › RT box -20C
- › 10X RT Random Primers
 - › RT box -20C
- › MultiScribe™ Reverse Transcriptase
 - › RT box -20C
- › Equipment
 - › Qubit
 - › Thermocycler
 - › Centrifuge
- › Primers per target
 - › Forward Primer (100 uM)
 - › Reverse Primer (100 uM)
- › SYBR Green (2X)
 - › 4C shelf (in a box)(bottomish shelf)
- › PCR Plates and seals
 - › Above qubit in common reagents cabinet
 - › Black plate with white tubes

Procedure

Prep

Program thermocycler program (can find under "fia" on controller 1)

Choose PCR program

Harvest Cells

Pipette mix and transfer conditions to eppendorf tubes or collection plate

Spin down at 400 xg for 5 min at 22C

Remove supernatant

Wash: Resuspend in 200 uL PBS, spin down at 300 xg for 5 min at 22C and remove supernatant

RNA Isolation

1. Isolate RNA using [RNAEasy kit](#), includes DNase digest

To optimize:

Warm DPEC water for elution at 55C

Incubate elution step for 10 min

All steps at RT and rather fast

Double elution - add eluted volume back through elution step?

2. Use a quantification method (ex. **Qubit**, Bioanalyzer, TapeStation)

3. Measure and bring RNA concentration to 100 ng/uL (> 105,000 cells)

Can normalize at a lower amount

Use up to 2 ug of total RNA per 20 uL reaction

Want 10 uL of RNA

Reverse Transcription

4. Allow kit components to thaw on ice

5. Prepare 2x cDNA reverse transcription master mix per sample

2x cDNA reverse transcription master mix						^
	A	B	C	D	E	
1		Final conc	Original conc	Volume (uL)	x 29	
2	10X RT Buffer	2X	10X	2.0	58	
3	dNTP Mix (100 mM)	2X	25X	0.8	23.2	
4	10X RT Random Primers	2X	10X	2.0	58	
5	MultiScribe™ Reverse Transcriptase	-	-	1.0	29	
6	RNase Inhibitor	-	-	0	0	
7	DPEC H2O	-	-	4.2	121.8	
8			Total	10	290	

6. Place on ice and mix gently

7. Pipette 10 µL of 2X RT master mix into each well of a 96-well reaction plate or individual tube.

8. Pipette 10 µL of RNA sample into each well, pipetting up and down two times to mix

9. Seal the plates or tubes.

10. Briefly centrifuge the plate or tubes to spin down the contents and to eliminate any air bubbles

(e.g., 1000 xg for 1 min)

11. Place the plate or tubes on ice until you are ready to load the thermal cycler

Run program: qPCR cDNA syn

Resuspend primers at this time or eat

qPCR cDNA syn			^
	A	B	
1	Temperature	Time	
2	25°C	10 min	
3	37°C	120 min	
4	85°C	5 min	
5	4°C	Forever	

12. Add 180 uL DPEC H2O

qPCR

13. Prepare a 10 uM dilution mix of forward + reverse primers for each target gene (x5 in this case)

Primer mix				^
	A	B	C	
1	Reagent	Volume (μl)	x 1	
2	Forward Primer (100 uM)	10	10	
3	Reverse Primer (100 uM)	10	10	
4	DPEC H2O	80	80	
5	Total	100	100	

14. Prepare master mix for each primer (total wells = 96) (always run triplicates)

GFI1B primers (12 x 3 = 24) --> 26 MM

all conditions get HKG:

Beta actin primers (24) --> 26 MM

18S primers (24) --> 26 MM

SYBR MM (*target*)								^
	A	B	C	D	E	F	G	
1	Reagent	Final Conc	OG conc	μl/well	x 3	x 10	x 31.5	
2	SYBR Green (2X)	~1X	2X	5	15	50	157.5	
3	DPEC H2O	-	-	3	9	30	94.5	
4	Diluted Primer Mix	1 μM	10 μM	1	3	10	31.5	
5	Total			9	27	90	283.5	

15. Add 9 ul SYBR MM per well

16. Add 1 ul diluted cDNA to each well

17. Vortex lightly or pipette mix

18. Seal the plate

19. Spin the plate briefly (e.g., 1000 xg for 1 min)

20. Run program in QPCR machine

qPCR Program Details			^
	A	B	
1	Temp	Time	
2	95C	5 min	
3	For 40 cycles		
4	95C	15 sec	
5	60C	30 sec	

Press new protocol and insert details

Then go to plate setup. First choose fluorophore but it is automatically set to SYBR. Then choose sample type (ex. unknown) then you can label them based on target and biological group. You can also label them based on replicates. Save plate setup and you should be ready to go. Open the machine using the computer "open" button, add your plate, and "close". Press run when you are ready.